

Residual-Shape Calibration and Empirical Drift Construction: Companion Note

Abstract

This note accompanies *Optimal Sizing of Leveraged Crypto Long–Short Positions under Kou Jump-Diffusion via Killed Moments*. It documents the empirical construction used in the paper’s showcase. A normalized, causal exponentially weighted mean (EWM) separates a horizon-matched directional signal from high-frequency innovations. A penalized empirical-characteristic-function (ECF) estimator then fits only the eleven diffusion, jump, and dependence parameters of a residual law normalized to zero expected log drift. The endpoint EWM signal, the residual-law price-growth correction, and role-specific lending or borrowing rates determine the final expected-price growth exponents and the long–short orientation.

Model notation

We retain the notation of the main paper. For assets $i = 1, 2$, the shape vector fitted to the residual returns is

$$\vartheta = ((\sigma_i, \lambda_i, p_i, \eta_{i,+}, \eta_{i,-})_{i=1,2}, \rho) \in \mathbb{R}^{11}.$$

Here $\sigma_i > 0$ is diffusion volatility, $\lambda_i > 0$ is jump intensity, $p_i \in (0, 1)$ is the upward-jump probability, $\eta_{i,+} \in (0, 1)$ and $\eta_{i,-} > 0$ are the mean upward and downward jump sizes, and $|\rho| < 1$ is Brownian correlation. Define the mean log jump, jump characteristic function, and price-jump compensator by

$$\bar{J}_i = p_i \eta_{i,+} - (1 - p_i) \eta_{i,-}, \quad \varphi_{J_i}(u) = \frac{p_i}{1 - iu\eta_{i,+}} + \frac{1 - p_i}{1 + iu\eta_{i,-}}, \quad \chi_i = \frac{p_i}{1 - \eta_{i,+}} + \frac{1 - p_i}{1 + \eta_{i,-}} - 1,$$

respectively. The ECF residual law is normalized to zero expected log drift. In the expected-price convention of the main paper this determines, rather than estimates,

$$\mu_i^{\text{res}} = \frac{1}{2} \sigma_i^2 + \lambda_i (\chi_i - \bar{J}_i), \quad \mu_i^{X, \text{res}} = -\lambda_i \bar{J}_i. \quad (1)$$

The residual characteristic exponent is therefore

$$\Psi_{\vartheta}^{\text{res}}(u, v) = i(-\lambda_1 \bar{J}_1 u - \lambda_2 \bar{J}_2 v) - \frac{1}{2} (\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho \sigma_1 \sigma_2 uv) + \lambda_1 (\varphi_{J_1}(u) - 1) + \lambda_2 (\varphi_{J_2}(v) - 1).$$

The liquidation-driving log-price spread is $Z = X_1 - X_2$. If moments through order K are required, the admissibility restriction is $K\eta_{i,+} < 1$ for $i = 1, 2$.

1. Causal EWM filtering and empirical drift

Let $r_{i,n}$ be the n th four-hour log return and let Δ be the corresponding year fraction. For the empirical horizon $T = 1/12$, set

$$\beta = 2^{-\Delta/T}$$

and form the finite-sample normalized EWM

$$m_{i,n} = \frac{\sum_{j=1}^n \beta^{n-j} r_{i,j}}{\sum_{j=1}^n \beta^{n-j}}. \quad (2)$$

Normalizing the weights avoids dependence on an arbitrary recursive starting value. The innovation used for shape calibration at time $n \geq 2$ is causal,

$$e_{i,n} = r_{i,n} - m_{i,n-1}, \quad \tilde{e}_{i,n} = e_{i,n} - \frac{1}{N-1} \sum_{\ell=2}^N e_{i,\ell}. \quad (3)$$

Thus the current return does not enter its own predictor, and the innovations supplied to the ECF estimator have sample mean exactly zero. The 540 raw WETH/WBTC return pairs yield 539 centered innovations. Here $T/\Delta = 182.5$ four-hour observations, $\beta = 0.9962091$, and the equivalent pandas-style span is 526.584.

Exponential-smoothing choices generally depend on the forecast horizon (Cox, 1961; Tiao and Xu, 1993). The half-life- T rule is used as a transparent horizon-matching convention, not as an estimated optimum. The sample contains only about three non-overlapping one-month forecast horizons, which is insufficient for a persuasive out-of-sample choice among decay rates. Moreover, high-frequency sampling is much more informative about volatility than expected return (Merton, 1980). These considerations motivate keeping directional drift outside the ECF fit. Sensitivity to the drift signal and decay horizon is consequently a decision-stage issue in the main paper.

At the sample endpoint, define the annualized EWM log-drift signal

$$d_i^{\text{EWM}} = m_{i,N}/\Delta$$

and the pre-financing price-growth exponent

$$v_i = d_i^{\text{EWM}} + \mu_i^{\text{res}}. \quad (4)$$

For each candidate ordered pair (L, S) of distinct assets, apply the sample-average rates over the observation window associated with the candidate roles and compute

$$\mu_L^{\text{emp}}(L, S) = v_L + r_{\text{supply}, L}, \quad \mu_S^{\text{emp}}(L, S) = v_S + r_{\text{borrow}, S}, \quad \Delta_\mu(L, S) = \mu_L^{\text{emp}}(L, S) - \mu_S^{\text{emp}}(L, S). \quad (5)$$

The selected orientation is the candidate with the largest positive $\Delta_\mu(L, S)$. Thus financing enters the orientation rule itself; v_i alone does not determine which asset is long. If no candidate has positive score, the rule does not produce a showcase trade.

2. Penalized ECF estimator

The closed-form residual Lévy–Khintchine exponent $\Psi_\vartheta^{\text{res}}(u, v)$ gives the model-implied characteristic function

$$\phi_\vartheta^{\text{res}}(u, v; \Delta) = \exp(\Delta \Psi_\vartheta^{\text{res}}(u, v)), \quad (6)$$

where Δ is the observation interval. Given N_{res} centered innovation pairs $(\tilde{e}_{1,n}, \tilde{e}_{2,n})_{n=1}^{N_{\text{res}}}$, the empirical characteristic function is

$$\hat{\phi}_{N_{\text{res}}}(u, v) = \frac{1}{N_{\text{res}}} \sum_{n=1}^{N_{\text{res}}} \exp(i(u\tilde{e}_{1,n} + v\tilde{e}_{2,n})). \quad (7)$$

The ECF component of the estimator is the normalized weighted squared distance between (7) and (6) over a discrete frequency grid $\mathcal{G}$:

$$Q_{N_{\text{res}}}^{\text{ECF}}(\vartheta) = \frac{\sum_{(u,v) \in \mathcal{G}} w(u,v) \left| \widehat{\phi}_{N_{\text{res}}}(u,v) - \phi_{\vartheta}^{\text{res}}(u,v;\Delta) \right|^2}{\sum_{(u,v) \in \mathcal{G}} w(u,v)}. \quad (8)$$

Because finite high-frequency samples leave a pronounced intensity–jump-size ridge, the reported calibration uses two transparent soft anchoring terms:

$$\begin{aligned} \widehat{\vartheta} &= \arg \min_{\vartheta \in \mathcal{V}} \mathcal{Q}_{N_{\text{res}}}(\vartheta), \\ \mathcal{Q}_{N_{\text{res}}}(\vartheta) &= Q_{N_{\text{res}}}^{\text{ECF}}(\vartheta) + \omega_{\lambda} \sum_{i=1}^2 (\log \lambda_i - \log \widetilde{\lambda}_i)^2 \\ &\quad + \omega_{\eta} \sum_{i=1}^2 \sum_{s \in \{+, -\}} (\log \eta_{i,s} - \log \widetilde{\eta}_{i,s})^2. \end{aligned} \quad (9)$$

Here $\widetilde{\lambda}_i$ is the variance/fourth-cumulant initializer and $\widetilde{\eta}_{i,s}$ is the peak-over-threshold (POT) mean-excess initializer described below. The application uses $(\omega_{\lambda}, \omega_{\eta}) = (0.05, 0.02)$. These are soft penalties, not equality restrictions; setting both weights to zero recovers pure ECF matching. This is the empirical-characteristic-function approach of Singleton (2001), which is particularly natural for affine and Lévy-based asset-pricing models whose characteristic functions are known in closed form. Feuerverger and McDunnough (1981) establish that ECF estimators over a rich grid can attain maximum-likelihood efficiency. The procedure avoids evaluating the bivariate transition density, which would require a two-dimensional numerical Fourier inversion inside every optimizer call. For background on DEJD likelihood estimation in the univariate case see Ramezani and Zeng (2007).

3. Frequency grid

The shape vector ϑ has the eleven free components stated above; neither μ_1 nor μ_2 is an ECF coordinate. Raw return magnitudes depend on the sampling interval, so fixed raw Fourier frequencies are poorly portable. Let s_1, s_2 , and s_Z be IQR-based robust scales, $s = \max\{(q_{0.75} - q_{0.25})/1.349, 10^{-8}\}$, for $\widetilde{e}_1, \widetilde{e}_2$, and $\widetilde{e}_1 - \widetilde{e}_2$. Dimensionless base frequencies are mapped to raw frequencies by division by the corresponding scale. The grid contains the following groups, with every point accompanied by its conjugate $(-u, -v)$:

- marginal points $(a/s_1, 0)$ and $(0, a/s_2)$ for $a \in \mathcal{A}_{\text{marg}}$;
- same-sign and mixed-sign joint points $(a/s_1, b/s_2)$ and $(a/s_1, -b/s_2)$ for $a, b \in \mathcal{A}_{\text{joint}}$, which are informative about ρ ;
- spread points $(a/s_Z, -a/s_Z)$ for $a \in \mathcal{A}_Z$, which directly match the log-ratio process governing liquidation; and
- high-base marginal and spread points for $a \in \mathcal{A}_{\text{tail}}$, included to retain information about jump-tail shape.

The dimensionless base nodes are

$$\begin{aligned}\mathcal{A}_{\text{marg}} &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5, 7.5, 10\}, \\ \mathcal{A}_{\text{joint}} &= \{0.25, 0.5, 1, 2, 3, 5\}, \\ \mathcal{A}_Z &= \{0.1, 0.2, 0.5, 0.75, 1, 1.5, 2, 3, 5\}, \\ \mathcal{A}_{\text{tail}} &= \{8, 12, 16, 20, 25\}.\end{aligned}$$

Points whose absolute raw coordinate exceeds 5,000 are discarded to avoid numerical aliasing. Including conjugate pairs, the unfiltered construction has 236 frequency evaluations; this cap leaves 234 for the application sample. Weights depend on dimensionless rather than raw frequency:

$$w(a, b) = \exp\{-0.02(a^2 + b^2)\}.$$

The ordinary spread group receives an additional factor of two. Thus the criterion emphasizes the barrier-driving ratio without making the choice of observation interval determine the effective frequency weights.

4. Reparameterisation and initialisation

Unconstrained reparameterisation. The parameter constraints stated above are enforced automatically via the change of variables

$$\sigma_i = e^{a_i}, \quad \lambda_i = e^{b_i}, \quad p_i = \text{sigmoid}(c_i), \quad \eta_{i,+} = \eta_{i,+}^{\max} \text{sigmoid}(d_i), \quad \eta_{i,-} = e^{e_i}, \quad \rho = \tanh(f), \quad (10)$$

where $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$. The optimized vector is the eleven-dimensional shape coordinate

$$\tau_{\text{shape}} = (a_1, b_1, c_1, d_1, e_1, a_2, b_2, c_2, d_2, e_2, f).$$

There is no drift coordinate: at every objective evaluation the two residual exponents are reconstructed from the candidate shape by (1). The optimization uses box bounds in the transformed space. For both assets the code uses a scaled sigmoid with

$$\eta_{i,+}^{\max} \leq (1 - \varepsilon)/K, \quad i = 1, 2,$$

so that $K\eta_{i,+} < 1$ is enforced for $i = 1, 2$, ensuring finiteness of the exponential moments used through order K .

Threshold initialiser. The characteristic-function objective is non-linear and non-convex, so the optimisation is initialised by a threshold-based decomposition following the logic of Mancini (2009) for discretely observed jump-diffusions. Let $q_i = \text{median}(\tilde{e}_{i,n})$ and let s_i be the IQR robust scale. For a frequency-aware threshold constant $c \in [3, 5]$, define candidate jump observations

$$\mathcal{J}_i = \{n : |\tilde{e}_{i,n} - q_i| > cs_i\}, \quad \mathcal{C}_i = \{1, \dots, N_{\text{res}}\} \setminus \mathcal{J}_i.$$

Initial volatility and correlation values use the continuous subset $\mathcal{C}_i$, while the jump-size means are anchored by peak-over-threshold excess means and the intensities are corrected by a variance/fourth-cumulant estimate. Schematically:

$$\begin{aligned}p_i^{(0)} &= \frac{|\mathcal{J}_{i,+}|}{|\mathcal{J}_i|} \quad \text{clamped to the admissible probability range,} \\ \eta_{i,+}^{(0)}, \eta_{i,-}^{(0)} &= \text{POT mean-excess estimates,} \\ \lambda_i^{(0)} &= \text{two-moment estimate from variance and fourth cumulant,} \\ \sigma_i^{(0)} &= \text{sd}(\tilde{e}_{i,n} : n \in \mathcal{C}_i) / \sqrt{\Delta}, \quad \rho^{(0)} = \text{corr}(\tilde{e}_1, \tilde{e}_2)_{n \in \mathcal{C}_1 \cap \mathcal{C}_2}.\end{aligned}$$

No sample-mean or drift initializer enters the optimization; the residual drift is imposed by (1). The threshold filter is not the final estimator. It is a warm start that places the optimizer in a sensible region of parameter space; see the remark in Mancini (2009).

Multi-start cloud. Because $\mathcal{Q}_{N_{\text{res}}}(\tau_{\text{shape}})$ may have local minima, the optimizer is run from a structured cloud. It contains the unperturbed initializer and six deterministic intensity–size ridge starts with

$$\lambda_i \mapsto c\lambda_i, \quad \eta_{i,\pm} \mapsto \eta_{i,\pm}/\sqrt{c}, \quad c \in \{0.1, 0.25, 0.5, 2, 4, 8\}.$$

Remaining starts cycle through four high-jump/high-diffusion scenarios and add bounded random perturbations to scale, probability, and correlation parameters. The application calibration uses 30 starts (the library default is 20) and returns the bounded L-BFGS-B minimizer with the lowest penalized objective.

5. Empirical showcase output

The application uses four-hour observations from 2026-02-26 to 2026-05-27 and calibrates the residual shape in the market-data order WETH/WBTC. The shape-only optimization converged with zero-based best-start index 2 among 30 starts, using 234 Fourier points, with penalized objective 0.00379935. Re-ordering the fitted law after comparing the two role-adjusted candidate assignments gives the shape parameters in Table 1.

Parameter	WBTC	WETH
σ_i	0.256199	0.313328
λ_i	321.411575	609.836386
p_i	0.497268	0.427636
$\eta_{i,+}$	0.010556	0.011327
$\eta_{i,-}$	0.004586	0.005590

Table 1: Penalized shape-only ECF estimates, reordered to the selected long/short asset order. The fitted Brownian correlation is $\rho = 0.841739$.

Table 3 records the separate drift construction. The endpoint EWM signals are 0.095089 for WBTC and -0.555894 for WETH. Adding the shape-implied zero-log-mean corrections gives pre-financing exponents 0.149288 and -0.462118 , respectively. Table 2 enumerates both ordered assignments after applying the candidate-specific supply and borrowing rates. Long WBTC/short WETH has the maximal positive score, 0.583607, whereas the reverse assignment has negative score.

Candidate long L	Candidate short S	$\mu_L^{\text{emp}}(L, S)$	$\mu_S^{\text{emp}}(L, S)$	$\Delta_\mu(L, S)$
WBTC	WETH	0.149344	-0.434263	0.583607
WETH	WBTC	-0.440217	0.152768	-0.592985

Table 2: Enumeration of the two role-adjusted candidate assignments. The rule selects the candidate with the largest positive drift spread, hence long WBTC and short WETH.

Annualized component	Long WBTC	Short WETH
Endpoint EWM log drift d_i^{EWM}	0.095089	−0.555894
Residual price-growth correction μ_i^{res}	0.054199	0.093776
Pre-financing exponent v_i	0.149288	−0.462118
Role-specific supply/borrowing rate	0.0000556	0.027855
Final price-growth exponent μ_i^{emp}	0.149344	−0.434263

Table 3: Empirical drift construction for the selected long–short orientation. The residual corrections are determined by the fitted shapes; they are not free ECF drift estimates.

6. Identifiability and model-comparison protocol

The penalized shape criterion $\mathcal{Q}_{N_{\text{res}}}(\vartheta)$ is non-convex. Jump-diffusion models possess approximate ridges along directions that preserve the compound-Poisson variance contribution $\lambda_i \mathbb{E}[J_i^2] = 2\lambda_i(p_i\eta_{i,+}^2 + (1-p_i)\eta_{i,-}^2)$ (Ramezani and Zeng, 2007). The diffusion volatilities σ_i and the Brownian correlation ρ are generally better identified from short samples than jump-size and intensity parameters, which carry larger finite-sample uncertainty unless the observation window contains many jump events. Because λ_i is annualized, a one-year window contains λ_i jumps in expectation, irrespective of whether prices are sampled hourly or daily. Finer sampling can make individual jumps easier to separate from aggregated returns, but it does not multiply the underlying annual Poisson count.

For descriptive fit comparisons on a common grid, compare the unpenalized component $Q_{N_{\text{res}}}^{\text{ECF}}(\hat{\vartheta})$ across nested residual laws (Gaussian, Merton, bivariate Kou), rather than the penalized values $\mathcal{Q}_{N_{\text{res}}}(\hat{\vartheta})$ whose anchors differ. Such comparisons should be accompanied by liquidation-relevant diagnostics: empirical versus model Z-tail quantiles, empirical versus model barrier-crossing frequencies, and simulated versus observed drawdowns of Z . These are more directly informative for the sizing application than standard likelihood-based criteria. The present empirical illustration does not carry out a formal GBM–Merton–Kou selection exercise. Its point calibration and sizing results remain conditional on the selected Kou model and stated empirical choices; they are not evidence of model dominance or parameter-robust recommendations.

The repository’s calibration package contains the implementation. The synthetic calibration job tests recovery of the eleven residual-shape parameters under the same zero-expected-log-drift restriction and produces Figure 1.

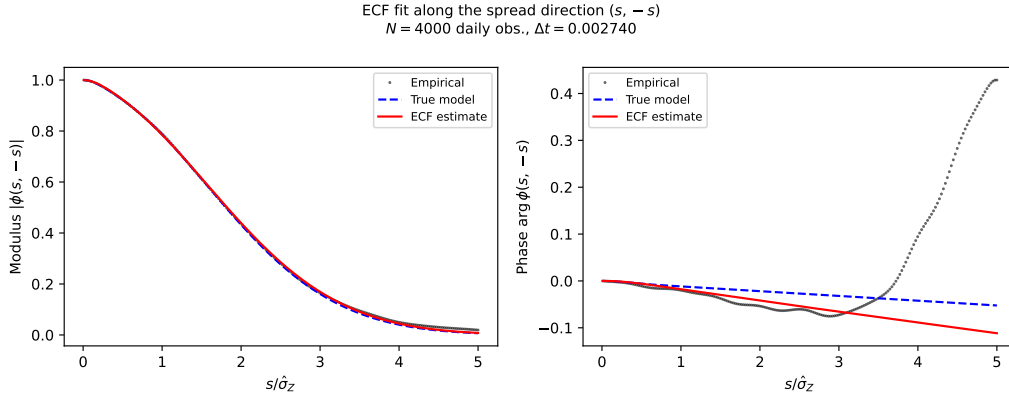


Figure 1: Shape-only ECF fit along the spread direction $(s, -s)$ for $N = 4,000$ synthetic daily observations from a residual law with zero expected log drift. Left: modulus $|\phi(s, -s)|$; right: phase $\arg \phi(s, -s)$. Dots: empirical CF $\hat{\phi}_N$; dashed: true residual-law CF; solid red: shape-only ECF estimate. Both residual expected log drifts are imposed at zero, and the corresponding price-growth corrections are derived from the shapes rather than estimated as drift coordinates. The close agreement in modulus and phase documents the fit along the liquidation-relevant spread direction; this plot alone does not assess every component of the full-grid criterion or validate an empirical directional-drift forecast. Script `optimal_long_short/job_runners/calibrate_kou.py`.

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